Question 4 : Matrices, Graphs & Trees.

a) $A = \begin{bmatrix} 2 & 5 \\ 4 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 8 & 10 & 1 \\ 7 & 4 & 1 \end{bmatrix}$

i) Identity Matrix of A.

A is a 2×2 Matrix, Therefore its Identity Matrix is also 2×2.

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ones on the diagonal zero's everywhere else.

always denoted with I (Identity)

ii) evaluate $2B - AB$.

First get $2B$ (multiply Matrix B by 2)

$$2B = 2\begin{bmatrix} 8 & 10 & 1 \\ 7 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 16 & 20 & 2 \\ 14 & 8 & 2 \end{bmatrix}$$

Matrix Multiplication.

$$AB = \begin{bmatrix} 2 & 5 \\ 4 & 1 \end{bmatrix}\begin{bmatrix} 8 & 10 & 1 \\ 7 & 4 & 1 \end{bmatrix}$$

number of cols in A must equal the number of rows in B.

2×2 2×3

size of ANS Matrix

$$= \begin{bmatrix} (2)(8)+(5)(7) & (2)(10)+(5)(4) & (2)(1)+(5)(1) \\ (4)(8)+(1)(7) & (4)(10)+(1)(4) & (4)(1)+(1)(1) \end{bmatrix}$$

$$= \begin{bmatrix} 51 & 40 & 7 \\ 39 & 44 & 5 \end{bmatrix}$$

→ over

$$\therefore 2B - AB = \begin{bmatrix} 16 & 20 & 2 \\ 14 & 8 & 2 \end{bmatrix} - \begin{bmatrix} 51 & 40 & 7 \\ 39 & 44 & 5 \end{bmatrix}$$

$$2B - AB = \begin{bmatrix} -35 & -20 & -5 \\ -25 & -36 & -3 \end{bmatrix}$$

iii) Determinant & Inverse of A (2x2 matrix)

$$A = \begin{bmatrix} 2 & 5 \\ 4 & 1 \end{bmatrix} \qquad \det A = ad - bc$$

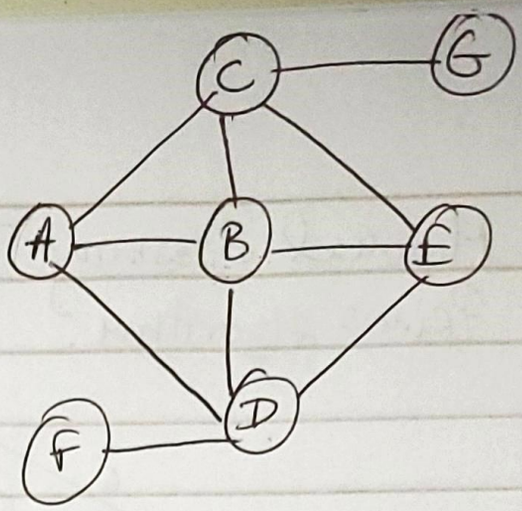
$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det A = (2)(1) - (5)(4)$$
$$= -18$$

$$A^{-1} = -\frac{1}{18} \begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{18} & \frac{5}{18} \\ \frac{4}{18} & -\frac{2}{18} \end{bmatrix} = \begin{bmatrix} -1/18 & 5/18 \\ 2/9 & -1/9 \end{bmatrix}$$

b) Graph Traversals.



i) Breadth first Search
uses a queue
data structure to help
keep track in the Algorithm.

Queue: ~~B~~ ~~C~~ ~~D~~ ~~E~~ ~~G~~ ~~F~~

BFS Output: ABCDEGF

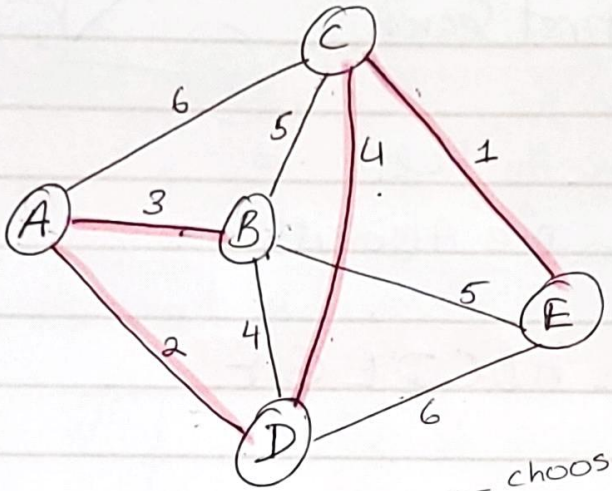
ii) Depth First Search
uses a stack data structure to keep
track in the algorithm.

~~G~~ ~~F~~ ~~D~~ ~~E~~ ~~C~~ ~~B~~
stack: ~~A~~

DFS Output: ABCEDFG

See notes for description, algorithm & example
of using each algorithm BFS & DFS..

c) Minimal Spanning Tree (MST) using Prim's Algorithm.



choos dc over db as db would form a cycle.

$T = \{ A\ ad\ D\ ab\ B\ dc\ C\ ce\ E \}$

do not have to go from the last vertex added, look at all vertices currently in T

5 vertices, therefore $5-1=4$ edges in the resulting MST.

Total weight $= 2+3+4+1 = 10$.